SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE CODE: CSA16   COURSE NAME: Data Warehousing and Data Mining

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COURSE OUTCOME COVERED

CO3: Evaluate the effectiveness of classification techniques on real-time data for accurate prediction, enabling informed decision-making. (BL3, BL4, BL5)
Assessment Tool Weightage: 10%

QUESTIONS: 30   **Total Marks:30**

Annexure A

MCQ TEST

Criteria	Conceptual Understanding *(3)*	Accuracy of Answers *(2.5)*	Application of Knowledge *(2)*	Time Management *(1.5)*	Question Interpretation *(1)*	*Total(10)*
Weightage	*30%*	25%	20%	15%	10%	*100%*
Q						

Code of Conduct:

I Haresh Kumar N L (192425009) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding; majority of answers incorrect
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions

ANNEXURE A

MCQ Test:

The Answers Are Highlighted in Yellow

1.Consider a simple linear regression model with one independent variable. Which of the following is true? (CO3)
Option_a: The regression line always passes through the mean of X and Y.
Option_b: The slope is always positive.
Option_c: The intercept is always positive.
Option_d: The correlation coefficient is always 1.

2.Which of the following statements is correct about Decision Trees? (CO3)
Option_a: Decision trees can only handle numerical data.
Option_b: Decision trees are always prone to overfitting.
Option_c: Decision trees can be used for classification and regression.
Option_d: None of the above.

3.Which of the following methods can be used for pruning in decision trees? (CO3)
Option_a: Pre-pruning
Option_b: Post-pruning
Option_c: Both (a) and (b)
Option_d: None of the above.

4.Which of the following is a disadvantage of Decision Trees? (CO3)
Option_a: Prone to overfitting.
Option_b: Requires large datasets.

Option_c: Cannot handle categorical data.
Option_d: Always has a high accuracy.

5.What does Bayes' Theorem primarily help with in probability and statistics? (CO3)
Option_a: Determining prior probabilities only
Option_b: Updating probabilities based on new evidence
Option_c: Computing standard deviation of a dataset
Option_d: Finding mean and median in distributions

6.In Bayes' Theorem, what does the term P(A|B) represent? (CO3)
Option_a: The probability of event B given event A has occurred
Option_b: The probability of event A given event B has occurred
Option_c: The joint probability of A and B
Option_d: The marginal probability of B

7.A factory has two machines, A and B. Machine A produces 60% of the products, and machine B produces 40%. If 2% of A's products and 3% of B's products are defective, what is the probability that a randomly selected defective product came from Machine B? (CO3)
Option_a: 0.40
Option_b: 0.50
Option_c: 0.60
Option_d: 0.75

8.Which of the following is not required to apply Bayes' Theorem? (CO3)
Option_a: Prior probability of the event
Option_b: Conditional probability of the evidence
Option_c: Total number of possible outcomes
Option_d: Probability of the evidence

9.A doctor uses a test to diagnose a disease that affects 1% of a population. The test correctly identifies the disease 95% of the time but also gives a false positive in 5% of cases. If a patient tests positive, what is the probability they actually have the disease? (CO3)
Option_a: 0.16
Option_b: 0.34
Option_c: 0.95
Option_d: 0.50

10.Bayes' Theorem is most useful when dealing with which type of probability? (CO3)
Option_a: Independent events
Option_b: Conditional probability
Option_c: Mutually exclusive events
Option_d: Uniform probability distribution

11. If a hypothesis HHH has a prior probability of 0.3 and new evidence EEE has a likelihood of 0.8 given HHH, how does Bayes' Theorem adjust the probability of HHH after seeing EEE? (CO3)
Option_a: The probability of HHH increases
Option_b: The probability of HHH decreases
Option_c: The probability of HHH remains unchanged
Option_d: The probability of HHH is completely unknown

12.A bag contains 5 red and 7 blue balls. A ball is drawn randomly and found to be red. What is the probability that the next ball drawn is also red (assuming the first ball is not replaced)? (CO3)
Option_a: 5/12
Option_b: 4/11

Option_c: 5/11
Option_d: 6/12

13.Which field heavily relies on Bayes' Theorem for updating beliefs based on new data? (CO3)
Question 230: END
Option_a: Machine Learning
Option_b: Geometry
Option_c: Classical Physics
Option_d: Cryptography

14.Why is backpropagation crucial in training deep neural networks? (CO3)
Option_a: It prevents overfitting by pruning unnecessary weights.
Option_b: It updates weights using the gradient of the loss function.
Option_c: It ensures each layer has the same activation values.
Option_d: It removes redundant layers to optimize the model.

15.During backpropagation, what happens when the learning rate is too high? (CO3)
Option_a: The model converges faster with high accuracy.
Option_b: The model may overshoot the optimal weight values and fail to converge.
Option_c: The model will always get stuck in local minima.
Option_d: The number of hidden layers increases automatically.

16.What is the main advantage of rule-based induction models? (CO3)
Question 247: END
Option_a: They provide an interpretable set of rules for decision-making.
Option_b: They work well with both numerical and categorical data.
Option_c: They can generalize well without requiring large datasets.
Option_d: Both (a) and (b)

17. Which of the following is an example of a rule-based induction algorithm? (CO3)
Question 248: END
Option_a: Decision Tree
Option_b: Naïve Bayes
Option_c: Support Vector Machine
Option_d: K-Nearest Neighbors

18.In rule-based induction, what happens when the rules are too specific to the training data?
Option_a: The model may suffer from overfitting.
Option_b: The model will generalize better.
Option_c: The rules will be shorter and more efficient.
Option_d: The accuracy will always increase.

19.Which approach is commonly used to simplify rule-based induction models? (CO3)
Question 250: END
Option_a: Pruning redundant or less significant rules
Option_b: Adding more features to the dataset
Option_c: Increasing the depth of the decision tree
Option_d: Using more training samples

20. What is the primary purpose of a confusion matrix in evaluating a classifier? (CO3)
Option_a: To visualize the performance of a classifier by showing correct and incorrect predictions
Option_b: To calculate the computational complexity of the classifier
Option_c: To determine the training time of the classifier
Option_d: To identify the features used by the classifier

21.Which metric is used to measure the proportion of correctly classified instances out of the total instances? (CO3)
Question 252: END
Option_a: Precision
Option_b: Recall
Option_c: Accuracy
Option_d: F1-Score

22.In a binary classification problem, what does the True Positive (TP) value represent? (CO3)
Question 253: END
Option_a: The number of negative instances correctly classified as negative
Option_b: The number of positive instances correctly classified as positive
Option_c: The number of positive instances incorrectly classified as negative
Option_d: The number of negative instances incorrectly classified as positive

23.What is the formula for calculating precision in a classification problem? (CO3)
Option_a: Precision = TP / (TP + FP)
Option_b: Precision = TP / (TP + FN)
Option_c: Precision = (TP + TN) / (TP + TN + FP + FN)
Option_d: Precision = TP / (FP + FN)

24.Which of the following metrics is most useful when the classes are imbalanced? (CO3)
Option_a: Accuracy
Option_b: F1-Score
Option_c: Recall
Option_d: Specificity

25.What does a high recall value indicate in a classification model? (CO3)
Question 256: END
Option_a: The model has a low number of false positives
Option_b: The model has a low number of false negatives
Option_c: The model has a high number of true negatives
Option_d: The model has a high number of false positives
Correct_option: The model has a low number of false negatives

26.Which of the following is an example of a rule-based induction algorithm? (CO3)
Option_a: Decision Tree
Option_b: Naïve Bayes
Option_c: Support Vector Machine
Option_d: K-Nearest Neighbor

27.In rule-based induction, what happens when the rules are too specific to the training data? (CO3)
Option_a: The model may suffer from overfitting.
Option_b: The model will generalize better.
Option_c: The rules will be shorter and more efficient.
Option_d: The accuracy will always increase.

28.Which approach is commonly used to simplify rule-based induction models? (CO3)
Option_a: Pruning redundant or less significant rules
Option_b: Adding more features to the dataset
Option_c: Increasing the depth of the decision tree
Option_d: Using more training samples

29.What is the primary purpose of a confusion matrix in evaluating a classifier? (CO3)
Option_a: To visualize the performance of a classifier by showing correct and incorrect predictions
Option_b: To calculate the computational complexity of the classifier

Option_c: To determine the training time of the classifier
Option_d: To identify the features used by the classifier

30.Which metric is used to measure the proportion of correctly classified instances out of the total instances? (CO3)
Question 252: END
Option_a: Precision
Option_b: Recall
Option_c: Accuracy
Option_d: F1-Score